Satyam Awasthi

Researcher & Software Engineer

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Kanpur, IN · UCSB & IIT Kharagpur Alum

Professional Summary

Computer Science researcher and software engineer specializing in **Autonomous Decision-Making**, **Spatial AI**, **Computer Vision**, and **Human-Computer Interaction (HCI)**. Focused on developing deterministic state frameworks and intelligent agent architectures under tight real-world constraints.

Education

University of California, Santa Barbara — Santa Barbara, CA
M.S. in Computer Science — Sep 2021 – Jun 2023
Graduate Research focused on Spatial AI, AR/VR, and HCI at the Four Eyes Lab.

Indian Institute of Technology, Kharagpur — Kharagpur, India
B.Tech. in Electrical Engineering (Minor in Computer Science) — Jul 2016 – Jul 2020
Graduated with First Class Honors.

Skills

Programming Languages (Expert): Python, C++, Kotlin, Java, MATLAB, SQL
Frameworks & Core Tools (Advanced): Android/iOS Native SDKs, LangChain, Pydantic (JSON Schema), OpenCV, ROS (Robot Operating System), PyTorch
Methodologies & Core Research: Autonomous Decision-Making, Human-Computer Interaction (HCI), Spatial AI, Agentic Architectures, Multi-Modal Sensor Fusion, Finite State Automation

Publications

Eye Tracking Performance in Mobile Mixed Reality
IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2024.
Investigated low-latency mobile eye tracking performance profiles across modern mixed reality layouts. Supported by NSF Grant #2211784.

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion
IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), 2023.
Proposed unique calibration logic pipelines specifically optimized for eye tracking modalities within mixed-reality locomotion patterns. Supported by NSF Grant #2211784.

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
14th International Conference on COMmunication Systems & NETworkS (COMSNETS), 2022.
Created inertial tracking methods to decode architectural hand stroke motions into alphanumeric symbols on wrist-worn wearables.

Experience

Intuit Inc — Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage **agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc — Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for client-side rendering efficiency.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021
Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Agentic Architectures, Sequential Decision Models, and Applied ML

JitterNot
An LSTM-based Model Predictive Controller for Adaptive Video Streaming

- Developed an LSTM network for real-time throughput prediction, simplifying the closed-loop control system from MISO to SISO.
- Implemented an MPC controller leveraging Receding Horizon Control (RHC) principles to dynamically calculate optimal video resolution.

In Motion — HAR using DeepConvLSTM
Distributed Mobile Computing System for Human Activity Recognition

- Designed a client-server telemetry pipeline streaming high-frequency accelerometer and gyroscope data for rolling window inference.
- Implemented a DeepConvLSTM core architecture to model spatial-temporal features using TimeDistributed CNN layers.

Autonomous Ground Vehicle & Swarm Robotics
End-to-End Perception and Planning for Robotic Fleets

- Built an autonomous navigation stack in ROS, fusing heterogeneous sensor streams (LiDAR, IMU, GPS) via an Extended Kalman Filter (EKF).
- Implemented visual obstacle segmentation networks (SegNet/U-Net) along with global (A^*/Dijkstra) and local (DWA) path planners.
- Researched decentralized swarm coordination using peer-to-peer relative pose estimation (AprilTags) over an ad-hoc XBee mesh network.

CohesiveAR
Augmented Reality Framework for Texture Extraction and Homography Space Mapping

- Engineered an asset ingestion pipeline using Google ARCore and OpenCV to project 3D world coordinates into 2D display space.
- Implemented perspective wrap and homography transformations to rectify camera tilt skew into standardized orthogonal textures.

Teaching

University of California, Santa Barbara Santa Barbara, CA
Graduate Teaching Assistant Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).